

Assignment 1: Part B

Big Data Analysis

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Key features of the dataset

For the question, “Are females aged 15-45 with a certain blood group more prone to PCOS?”, each of the 15 variables in the dataset, both categorical and quantitative variables, show some level of association with the target variable, PCOS. These associations were determined using chi-square tests for independence¹ for categorical variables (Statistics Solutions n.d.). And point-biserial correlation² coefficients for quantitative variables (www.andrews.edu n.d.).

Table 1: Description of key features of the dataset.

Variable Name	Variable Type	Description
Age (yrs)	Quantitative	Age of individual.
Height (cm)	Quantitative	Height of the individual.
Weight (Kg)	Quantitative	Indicates the weight of the individual.
Cycle length (days)	Quantitative	The number of days an individual has menstruation in each cycle.
Weight gain	Categorical	Indicates whether the individual experienced weight gain (0,1).
BMI	Categorical	Indicates BMI category (Underweight, Normal Weight, Overweight, Obese)
Blood Group	Categorical	Indicates the blood group of the individual.
PCOS	Categorical	Indicates the presence of PCOS (0,1).
Regular cycle	Categorical	Indicates whether the individual has regular menstrual cycle (0,1).
Hair loss/baldness	Categorical	Indicates whether the individual has experienced hair loss or baldness (0,1).
Excessive body/facial hair	Categorical	Indicates the presence of excess hairgrowth (0,1).
Skin darkening	Categorical	Indicates presence of skin darkening in any body part (0,1).
Pimples on face/jawline	Categorical	Indicates experience of pimples (0,1).
Regular fast food intake	Categorical	Indicates if the individual regularly consumes fast food (0,1).
Regular physical activity	Categorical	Indicates if the individual does any regular physical activity (0,1).

¹ Chi square test for independence – To determine significant between two nominal (categorical) variables.

² Point-biserial coefficient - A special case of Pearson where one variable is quantitative and the other variable is dichotomous (values: 1 = presence, 0 = absence) and nominal.

Data analysis and visualizations

- Quantitative variables

Table 2: Summary statistics of numerical variables.

	Mean	Median	Quartile1	Quartile3	Mode	Standard Deviation	Variance	Skewness
Age (yrs)	30.05	30	26	35	28	6.41	41.15	0.293
Height (cm)	156.82	156.9	152.3	160.6	152	6.25	39.11	0.306
Weight (Kg)	59.52	58.5	52	65.7	55	11.38	129.58	0.577
Cycle length (days)	4.91	5	4	5	5	1.45	2.10	0.972

This summary offers a dataset overview to understand the distribution of data. Knowing the mean, median, quartiles, and skewness of age confirms the dataset predominantly represents younger half of the 15-45 age group. High variability in age, weight, and height suggests a diverse dataset. Positive skewness indicates comparatively more younger individuals, shorter heights, lower weights, and cycle lengths clustered around 5 days.

- Categorical variables

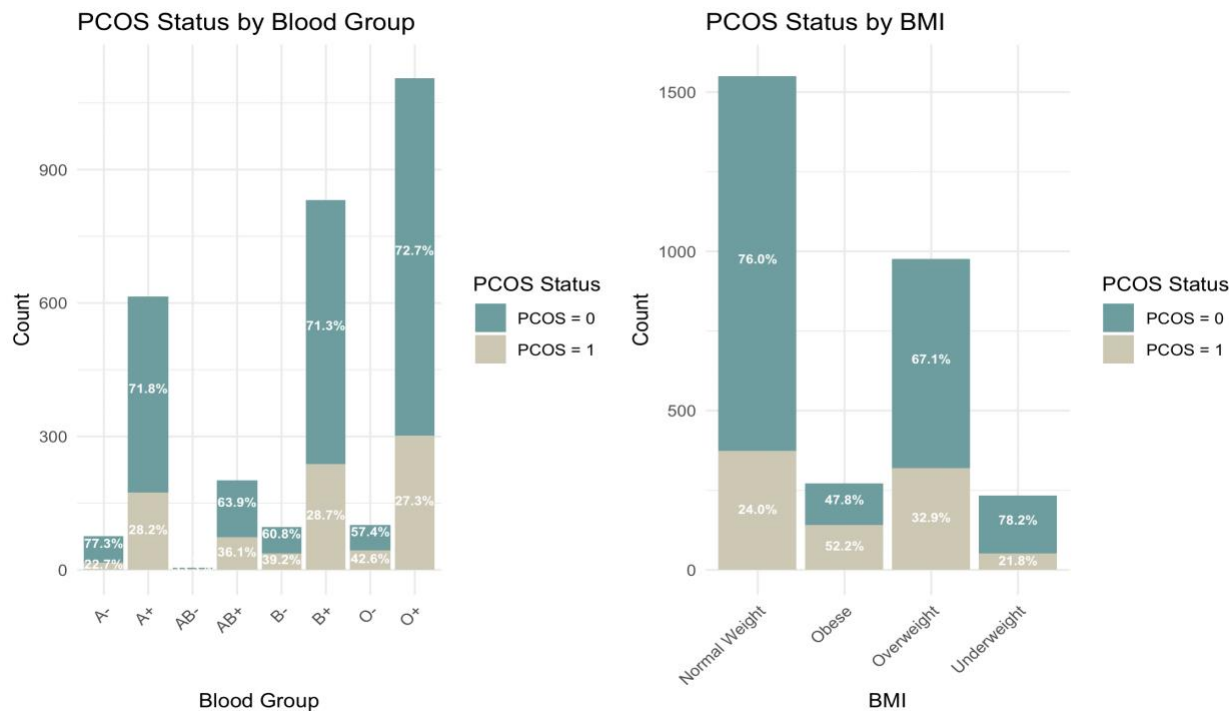


Fig 1: Stacked bar chart showing proportions of data who have PCOS (PCOS = 1) and who do not have PCOS (PCOS = 0) for each blood group and BMI category respectively.

The observation that individuals with blood group O+ and B+ have higher percentages of PCOS suggests a potential correlation between certain blood groups.

For BMI, the highest proportion of individuals with PCOS is observed among those categorized as obese. This indicates a strong association between obesity and the likelihood of having PCOS. It shows a clustering effect where the majority of individuals with PCOS are concentrated in the overweight and obese categories. This implies that analyzing BMI distributions across different blood types can identify higher-risk groups for PCOS within this demographic.

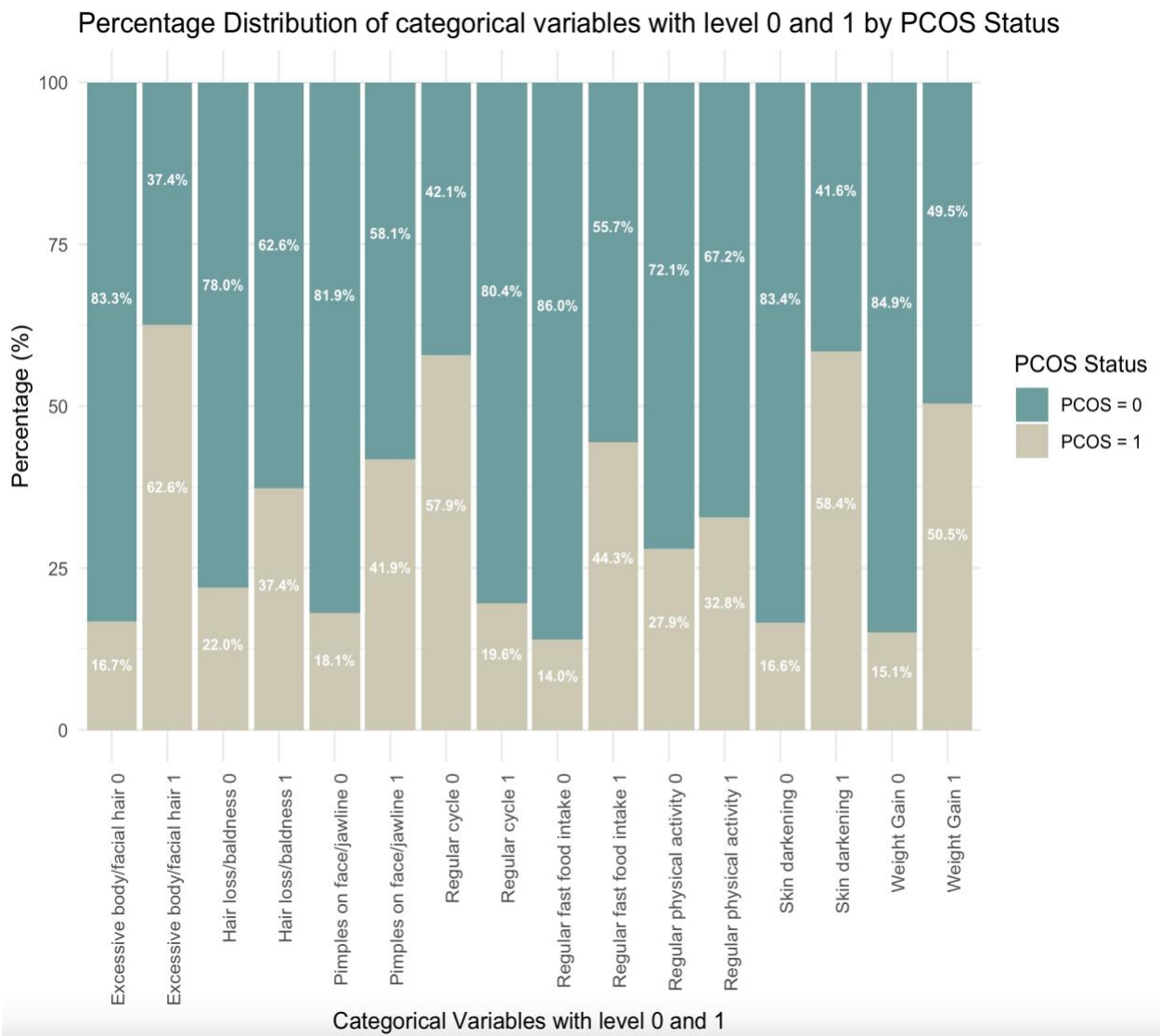


Fig 2: PCOS Status Distribution Across Categorical Variables depicting percentage for each categorical variable.

Individuals with PCOS exhibit higher rates of weight gain, irregular cycles, hair loss/baldness, excessive body/facial hair, pimples, and skin darkening. They also report more frequent fast food intake and less regular physical activity compared to those without PCOS. By accounting for these factors, we can better determine whether blood group independently affects the likelihood of developing PCOS, ensuring a more accurate assessment of its role.

Problem refinement and additional data source

The question does not need further refinement because it clearly defines the population of interest (females aged 15-45), and identifies clear independent (blood group) and dependent variables (PCOS status), allowing for straightforward and targeted statistical analysis.

Some additional data sources could enhance the analysis, such as:

- Family medical history related to any reproductive health conditions.
- Detailed diet and exercise routine and detailed hormonal profiles.
- Genetic data related to PCOS (Dutta 2019).

However, obtaining these additional sources is challenging because this data pertains to sensitive information about women's reproductive health and is not readily accessible.

The next steps will be to fit a model to investigate whether females ages 15-45 with certain blood types are more susceptible to PCOS compared to others. Then the performance of the model will be assessed using metrics such as accuracy, precision, recall and ROC curve.

References:

1. Datamation 2021, *What is Statistical Data Analysis? | Best Statistical Methods 2021*, viewed 20 June 2024, <<https://www.datamation.com/big-data/statistical-data-analysis/>>
2. Dutta, S.S 2019, 'Genetic of Polycystic Ovary Syndrome (PCOS)', *News-Medical.net*, November 7, viewed 26 June 2024, <[https://www.news-medical.net/health/Genetics-of-Polycystic-Ovary-Syndrome-\(PCOS\).aspx](https://www.news-medical.net/health/Genetics-of-Polycystic-Ovary-Syndrome-(PCOS).aspx)>
3. Rasquin, L and Mayrin, J.V 2022, *Polycystic Ovarian Disease (Stein-Leventhal Syndrome)*, Nih.gov, viewed 21 June 2024, <<https://www.ncbi.nlm.nih.gov/books/NBK459251/>>
4. Statistics Solutions n.d., *Chi-Square Test of Independence*, viewed 27 June 2024, <<https://www.statisticssolutions.com/free-resources/directory-of-statistical-analyses/chi-square/>>
5. www.andrews.edu (n.d.), *More Correlation Coefficients*, viewed 27 June 2024, <<https://www.andrews.edu/~calkins/math/edrm611/edrm13.htm>>